

Variance-Corrected Multi-Asset Equity Simulation with Hybrid Hidden Markov Marginals

Abstract

Synthetic multi-asset equity data must reproduce each asset’s return distribution and its relationship with the market. A common approach adds a synthetic full-return path to a market factor as though the synthetic path were a residual. Because that path already contains the asset’s total variance, the construction adds market variance a second time. We derived a correction that centers and rescales each synthetic path before adding the market factor. We tested the method on 423 non-market assets in a 424-asset US equity and exchange-traded-fund universe. The correction preserved each generator’s target variance, recovered its calibrated market loading with low error, and retained heavy-tailed return distributions without fitting a second model to the regression residuals. On 416 complete asset histories held out from 2025, it increased the mean Kolmogorov–Smirnov pass rate from 83.5% to 87.0% and reduced the median ratio of synthetic to observed variance from 1.29 to 0.97. Its one-day left-tail 99% Value-at-Risk exceedance rate was similar to that of the best residual-fit method, although both exceeded the nominal rate. We generated the per-asset paths with a hidden Markov model whose jump-duration variant lengthened visits to extreme-return states. On a decade of daily SPY data, this variant improved the match to volatility clustering relative to the no-jump model while maintaining strong distributional fit. We selected it as a practical compromise, not because it outperformed every baseline. The multi-asset method still omits residual dependence between assets, which can overstate the benefit of daily rebalancing. Code and fitted per-asset models are released with the paper.

Keywords: hidden Markov model, jump-duration mechanism, single-index model, synthetic financial data, variance correction, Value-at-Risk coverage check, volatility clustering, heavy-tailed marginals

1. Introduction

Synthetic financial data often fail to reproduce the statistical properties of real returns. These paths are used to stress-test risk models against unobserved market scenarios, validate portfolio optimization algorithms, and evaluate allocation strategies across many simulated scenarios (Assefa et al., 2020; Jordon et al., 2022). Empirical equity excess growth rates exhibit three statistical regularities a generative model must reproduce: a heavy-tailed marginal distribution, weak autocorrelation of raw growth rates, and persistent volatility clustering, in which large changes are followed by large changes and small changes by

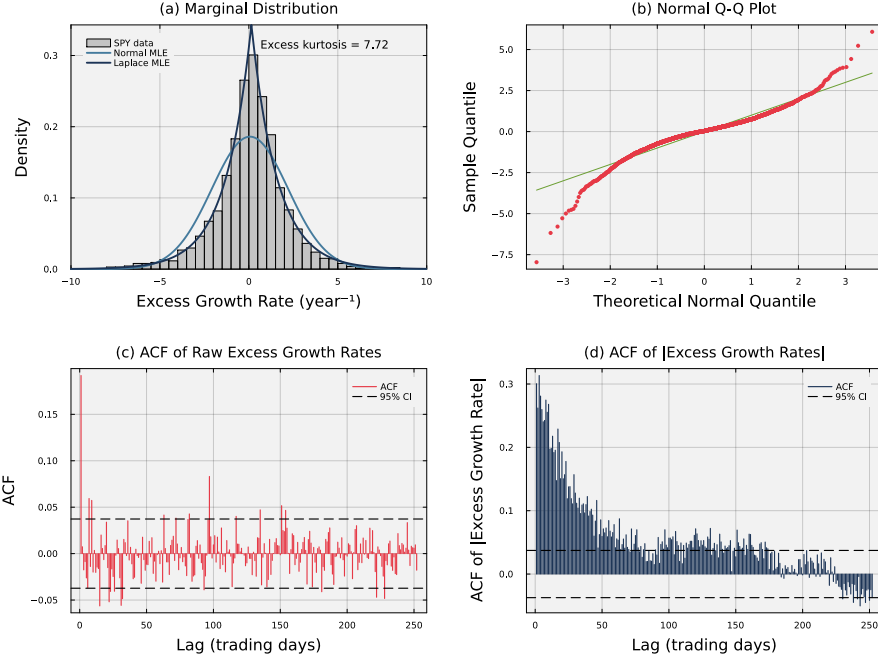


Figure 1: **Empirical stylized facts of SPY daily excess growth rates (2014–2024).** Panel (a) shows the heavy-tailed return density; the Laplace maximum-likelihood fit tracks the empirical peak and tails while the Gaussian fit does not. Panel (b) shows departure from the diagonal in a normal quantile-quantile plot at both extremes. Panel (c) shows the autocorrelation function of raw excess growth rates. Individual correlations are small, although the Ljung–Box test detects joint serial dependence. Panel (d) shows the autocorrelation function of absolute excess growth rates, which decays slowly across the reported lags.

small changes (Mandelbrot, 1963; Cont, 2001; Stenger et al., 2024). All three patterns are visible in SPY daily excess growth rates from 2014–2024 (Fig. 1). The empirical density is sharply concentrated near zero but places substantially more probability in the tails than a Gaussian model, as shown by the density and quantile-quantile comparisons (Fig. 1a and Fig. 1b). Raw growth rates have little serial correlation (Fig. 1c), whereas the positive, slowly decaying autocorrelation of their absolute values (Fig. 1d) shows that their magnitudes remain temporally dependent. These regularities are often called *stylized facts*. They show why matching the distribution of individual returns is not enough: raw returns should remain nearly uncorrelated, while the sizes of those returns should remain correlated over time.

Existing generators typically reproduce some of these regularities but not others. The generalized autoregressive conditional heteroskedasticity (GARCH) family (Engle, 1982; Bollerslev, 1986) captures volatility clustering but requires fat-tailed innovations to reproduce heavy tails. Stochastic-volatility and jump-diffusion models (Heston, 1993; Merton, 1976; Kou, 2002) produce heavy tails, but

the jump component itself does not create persistent volatility. Deep generative models (Takahashi et al., 2019; Yoon et al., 2019; Kwon and Lee, 2024) can learn complex marginals but often miss the absolute-return autocorrelation associated with volatility clustering. Standard hidden Markov models (HMMs) (Rabiner and Juang, 1986) provide a regime-switching structure but return to central regimes too quickly to dwell in tail states long enough to reproduce the empirical clustering profile (Rydén et al., 1998; Bulla and Bulla, 2006). We therefore sought a model that retained distributional fit, reduced absolute-return autocorrelation error, and could be fitted separately across a large asset universe.

We addressed the problem in two stages. First, we built a per-asset generator based on a hidden Markov model. It divided returns into Laplace-quantile states, used Student- t draws within each state, and estimated transitions by direct counting. Its jump-duration variant (HMM-WJ) added temporary episodes in the extreme-return states. On a decade of SPY data, HMM-WJ reduced the error in absolute-return autocorrelation from 0.059 to 0.053 relative to the no-jump model while retaining a 98.3% in-sample Kolmogorov–Smirnov pass rate. The GARCH model achieved a lower autocorrelation error of 0.031 but did not match the return distribution. We therefore used HMM-WJ as a practical per-asset generator, not as a uniformly superior model.

Our main contribution concerns the second stage: combining those per-asset paths with a shared market path. Treating a full synthetic return path as a single-index residual creates an unintended mean shift and counts market variance twice. We removed the path mean and derived a closed-form scale correction that preserved target variance and recovered the calibrated market loading across a 424-asset universe. With all 2014–2024 parameters held fixed, the correction increased the mean Kolmogorov–Smirnov pass rate from 83.5% to 87.0% on 416 complete asset histories from 2025 and reduced the median variance ratio from 1.29 to 0.97. The method does not, however, reproduce residual dependence between assets. Independent residuals created too much cross-sectional dispersion and could overstate the return from daily rebalancing. A location adjustment corrected the average return level but could not restore the missing dependence. This distinction between per-asset fit, variance preservation, and cross-asset dependence defines both the contribution and the limit of the method.

2. Related Work

Models of equity-market stylized facts follow either a bottom-up or top-down approach. Bottom-up agent-based models (Lux and Marchesi, 1999; Cont and Bouchaud, 2000; Farmer and Joshi, 2002; LeBaron, 2006; Arthur, 2021; Farmer, 2025) derive return dynamics from simulated interactions among heterogeneous agents. They are demanding to calibrate and expensive to simulate at scale. Top-down models instead fit statistical structures directly to observed growth-rate series. Top-down generators trace to Bachelier’s diffusion model, later formalized as geometric Brownian motion and used by Black-Scholes-Merton for option pricing (Bachelier, 1900; Osborne, 1959; Samuelson, 1965; Black and

Scholes, 1973). These diffusion baselines are tractable but fail on the empirical regularities that drive risk and pricing. Mandelbrot (1963) showed that returns have heavier tails than a Gaussian model allows. Fama (1970) reviewed evidence that raw returns have near-zero autocorrelation. Schwert (1989) documented persistence in monthly absolute residuals, a pattern later described by Cont (2001) as volatility clustering. The autoregressive conditional heteroskedasticity (ARCH) and generalized ARCH (GARCH) models of Engle (1982) and Bollerslev (1986) reproduced clustering but required a heavy-tailed noise term to produce heavy-tailed marginals. Neither model includes discrete regimes or sudden jumps. Stochastic-volatility models (Stein and Stein, 1991; Heston, 1993) make volatility a latent process to capture clustering, but estimating that process requires filtering or Markov chain Monte Carlo (MCMC) methods (Kim et al., 1998); jump-diffusion models (Merton, 1976; Kou, 2002) inject Poisson events into the price path, producing heavy tails by construction, but the jump component alone does not produce persistent volatility.

Hidden Markov models embed regime and jump behavior directly in a latent discrete chain. This approach follows the two-state regime-switching model of Hamilton (1989), later applied to volatility forecasting (Rossi and Gallo, 2006) and equity trading (Nguyen, 2018). Rydén et al. (1998) found that Gaussian-emission HMMs matched heavy tails and the absence of return autocorrelation but did not reproduce volatility clustering. Bulla and Bulla (2006) modeled clustering with hidden semi-Markov models (HSMMs) that replaced the geometric dwell-time distribution of every state with a parametric (negative-binomial) alternative. Likelihood-fitted HMM and HSMM formulations commonly use the Baum–Welch expectation-maximization (EM) algorithm (Baum et al., 1970), which is iterative and can be sensitive to initialization. Deep generative alternatives learn the marginal directly but can miss the volatility-clustering signature: Takahashi et al. (2019) found that generative adversarial network (GAN) architectures failed to reproduce the absolute-return autocorrelation function, and Kwon and Lee (2024) reported the same failure for TimeGAN (Yoon et al., 2019). Recent surveys (Assefa et al., 2020; Jordon et al., 2022; Stenger et al., 2024) agree that stylized-facts reproduction remains the primary quality criterion. Our marginal component also used a latent-state chain, but it differed from the HSMM. It kept the empirically counted transition matrix for the central states and added an event-driven override that forces the chain into the tail states for a Poisson-distributed duration on a subset of time steps.

Building a multi-asset path requires a dependence model in addition to the separate single-asset generators. Mixture hidden Markov models (Dias et al., 2010) share latent state across assets but are costly to estimate in high dimensions. Copula approaches (Embrechts et al., 2002; Cherubini et al., 2004; Aas et al., 2009) separate marginals from cross-asset dependence but require choosing a copula family and, in their standard static form, treat successive time slices as independent. Factor models instead reduce a firm’s growth rate to one or more shared factors (Ross, 1976; Fama and French, 1993). The single-index model (SIM) of Sharpe (1963) uses one shared factor and decomposes each asset’s growth rate into a market term and an idiosyncratic residual. This structure

can use a single-factor market generator directly. We derived a closed-form correction that preserves the generator variance while recovering the calibrated market loading.

A statistical match does not guarantee that synthetic paths will be useful in a downstream financial task. We therefore combined goodness-of-fit tests for the return distribution (Kolmogorov, 1933; Smirnov, 1948; Anderson and Darling, 1952) with autocorrelation diagnostics for temporal structure (Cont, 2001). We also evaluated Value-at-Risk coverage and a daily-rebalanced allocation example. Booth and Fama (1992) described the problem that arises in the latter setting: paths whose residuals are drawn independently across assets can create a *diversification return* that is muted in real markets, where common sector and style factors correlate residuals across assets. We measured Value-at-Risk performance with Kupiec’s unconditional-coverage test and used a location adjustment to measure, but not solve, the return-level effect of the missing residual dependence.

3. Methods

We fitted a marginal generator for each asset and then combined the generated paths with a variance-corrected single-index model. The dataset contained 424 United States-listed equities and exchange-traded funds observed from 2014 through 2024. The single-asset analysis focused on the SPDR S&P 500 ETF (SPY). It used 2,767 daily volume-weighted average prices for fitting, which yielded 2,766 growth rates, and the next 250 prices from 2025, which yielded 249 holdout growth rates. The multi-asset analysis used daily closing prices. Of the 423 non-market assets in the training universe, 416 had complete 2025 histories and entered the multi-asset holdout evaluation. We computed the excess growth rate $G_{i,j}$ for ticker i between trading days $j - 1$ and j from price $P_{i,j}$ as follows:

$$G_{i,j} \equiv \left(\frac{1}{\Delta t} \right) \ln \left(\frac{P_{i,j}}{P_{i,j-1}} \right) - r_f. \quad (1)$$

where $\Delta t = 1/252$ is the fraction of a trading year spanned by one trading day, and r_f is a constant continuously compounded risk-free rate. We used $r_f = 0.043$ for the SPY training period and $r_f = 0.0421$ for its 2025 holdout, based on Treasury STRIPS (Separate Trading of Registered Interest and Principal of Securities) bond yields. The excess growth rate has units of year^{-1} and is time-additive under continuous compounding.

3.1. Per-asset hidden Markov marginal generator

For each asset, HMM-WJ converted the continuous return series into a sequence of N ordered return regimes. At each time step, the model first selected a regime and then drew a return from that regime’s distribution (Fig. 2). We defined the model as follows: $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$ (Rabiner and Juang, 1986). The hidden state $S_t \in \mathcal{S} = \{1, \dots, N\}$ identifies the regime at time t , and the observed value is the excess growth rate G_t from Eq. (1). We suppress the ticker

index i when discussing one asset. The transition matrix $\mathbf{T}$ sets the probabilities of moving between regimes, $\mathbf{E}$ gives the return distribution within each regime, and $\bar{\pi}$ gives the initial regime probabilities.

We ordered the regimes from the most negative to the most positive returns. Their boundaries were equally spaced quantiles of a Laplace distribution fitted to the asset’s in-sample returns. Specifically, $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$, where μ_L and b_L are the fitted location and scale. We set $Q_0 = -\infty$ and $Q_N = +\infty$ and assigned return G_t to state k when $Q_{k-1} < G_t \leq Q_k$. The Laplace distribution captured the sharp concentration of small price movements in Fig. 1a (Kotz et al., 2001; Toth and Jones, 2019). Its closed-form quantiles also made the same partition inexpensive to compute across all 424 tickers (Bilmes, 1998).

The within-state emission followed a location-scale Student- t distribution:

$$G_t \mid S_t = k \sim \mu_k + \sigma_k \cdot t_\nu, \quad \nu = 5, \quad (2)$$

where t_ν denotes a standard Student- t random variable with ν degrees of freedom. For observations assigned to state k , μ_k is the sample mean and σ_k is the sample standard deviation. A standard t_ν has variance $\nu/(\nu - 2)$, so σ_k is a scale parameter rather than the emission standard deviation. At $\nu = 5$, that standard deviation is $\sigma_k \sqrt{5/3}$. We selected $\nu = 5$ from a sensitivity sweep ($\nu = \infty$ recovers the Normal baseline; explored range in Table 1). At $\nu = 5$, simulated kurtosis was 7.5 against the observed 7.7, compared with 5.5 under Normal emissions. The corresponding Student- t KS and AD pass rates were 98.3% and 94.8%. The full comparison against an HSMM-style baseline adapted from Bulla and Bulla (2006) is given in Supplementary Table S3 of Sec. S5.

With the regimes and within-regime distributions fixed, we estimated the transition matrix $\mathbf{T}$ by counting observed transitions. This avoided the initialization sensitivity of iterative expectation-maximization (Bilmes, 1998). The counted matrix was strongly concentrated near its diagonal, so a state usually persisted for only 1–2 days, including in the tails (Supplementary Fig. S5 in Online Appendix S4). As a result, the ordinary model left extreme-return states too quickly to reproduce the observed volatility clustering (Bulla and Bulla, 2006).

We addressed that problem with occasional jump episodes. Whenever no jump episode was active, a new episode began with probability ϵ . Its duration was $K \sim \text{Poisson}(\lambda)$, where λ is the mean episode length (Glasserman, 2003). Because a Poisson draw can be zero, $K = 0$ caused no forced tail step. For $K > 0$, the model temporarily replaced ordinary transitions with draws from one of two tail sets: $\mathcal{S}_- = \{1, \dots, N_{\text{tail}}\}$ for the most negative states and $\mathcal{S}_+ = \{N - N_{\text{tail}} + 1, \dots, N\}$ for the most positive states. It selected the negative set with probability p_{neg} and the positive set otherwise. The model made this choice independently at each forced step, allowing negative extreme returns to occur slightly more often than positive ones. Table 1 gives all four jump settings.

We obtained the initial-state distribution $\bar{\pi}$ by solving $\bar{\pi}\mathbf{T} = \bar{\pi}$ with $\sum_k \bar{\pi}_k = 1$ (Hamilton, 2018; Grinstead and Snell, 1997). This distribution is exact for the ordinary transition matrix but only approximate for the jump-augmented process

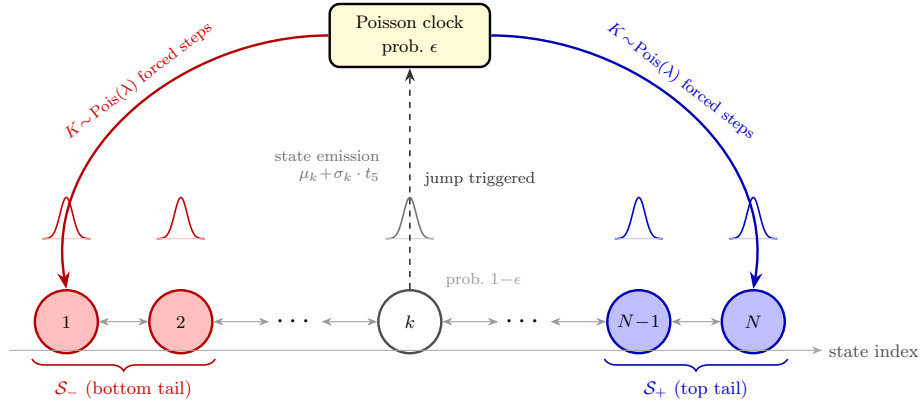


Figure 2: **Architecture of the hybrid hidden Markov model with a jump-duration mechanism (HMM-WJ).** When no jump episode is active, the chain transitions according to the empirical transition matrix $\mathbf{T}$ with probability $1 - \epsilon$ (thin bidirectional arrows) or enters a Poisson jump episode with probability ϵ (dashed upward arrow to the Poisson clock). During an episode, each of the $K \sim \text{Pois}(\lambda)$ forced steps independently draws the bottom tail set $\mathcal{S}_-$ (red, lowest-valued states) with probability p_{neg} or the top tail set $\mathcal{S}_+$ (blue, highest-valued states) otherwise, before normal Markovian evolution resumes. Small bell curves above each state depict the state-conditional location-scale Student- t ($\nu = 5$) emission distribution. Tail sets are defined as the N_{tail} lowest- and highest-quantile states under the Laplace quantile partition.

because it does not include the remaining jump duration. Algorithm 1 gives the full simulation procedure. Each step either followed $\mathbf{T}$ or continued a jump episode and then drew a return from the selected state's emission distribution.

We chose the jump probability ϵ and mean duration λ so that simulated paths matched two features of the SPY history: volatility clustering and kurtosis (Mandelbrot, 1963; Cont, 2001; Rydén et al., 1998). The jump mechanism follows the Poisson event idea used in jump-diffusion models (Merton, 1976; Kou, 2002), but here an event extends a visit to the tail states rather than adding an instantaneous price jump. We searched a grid of (ϵ, λ) values. For each jump-active path, the first part of the objective measured the squared difference between the observed and simulated autocorrelation functions of absolute returns through lag L . The second part measured the squared difference in excess kurtosis. We then averaged both errors over the jump-active paths. The calibration objective was:

$$J(\epsilon, \lambda) = \frac{1}{|\mathcal{R}_J|} \sum_{r \in \mathcal{R}_J} \left[\sum_{\tau=1}^L (\text{ACF}_{\text{obs}}(\tau) - \text{ACF}_r(\tau))^2 + w_\kappa (\kappa_{\text{obs}} - \kappa_r)^2 \right], \quad (3)$$

where $\mathcal{R}_J$ is the set of jump-active simulated paths, $\text{ACF}_{\text{obs}}(\tau)$ and κ_{obs} are the empirical absolute-growth autocorrelation at lag τ and excess kurtosis, and $\text{ACF}_r(\tau)$ and κ_r are the corresponding values for path r . We use κ for kurtosis to avoid collision with the jump-duration variable K . We simulated 200 independent paths of 2,766 trading days at each grid point. A path with no jump reduces to HMM-NJ and contains no information about (ϵ, λ) , so the objective included

Algorithm 1 HMM-WJ single asset marginal generator.

Require: Fitted parameters $(\mathbf{T}, \mathbf{E}, \bar{\pi}, \epsilon, \lambda, p_{\text{neg}}, N_{\text{tail}})$; horizon T ; tail sets $\mathcal{S}_- = \{1, \dots, N_{\text{tail}}\}$ and $\mathcal{S}_+ = \{N - N_{\text{tail}} + 1, \dots, N\}$.

Ensure: State sequence $\{S_t\}_{t=0}^T$ and growth-rate path $\{G_t\}_{t=1}^T$.

- 1: Draw $S_0 \sim \bar{\pi}$; $k \leftarrow 0$ { k : remaining forced-step counter}
- 2: **for** $t = 1, \dots, T$ **do**
- 3: **if** $k > 0$ **then**
- 4: $\mathcal{J} \leftarrow \mathcal{S}_-$ w.p. p_{neg} , else $\mathcal{S}_+$; $S_t \sim \text{Uniform}(\mathcal{J})$; $k \leftarrow k - 1$ {forced tail step; sign drawn per step}
- 5: **else if** $\text{Uniform}(0, 1) > \epsilon$ **then**
- 6: $S_t \sim \mathbf{T}_{S_{t-1}, \cdot}$ {standard Markov transition}
- 7: **else**
- 8: $K \sim \text{Poisson}(\lambda)$ {trigger jump; K is the episode length}
- 9: **if** $K = 0$ **then**
- 10: $S_t \sim \mathbf{T}_{S_{t-1}, \cdot}$ { $K = 0$: no forced steps, standard transition}
- 11: **else**
- 12: $\mathcal{J} \leftarrow \mathcal{S}_-$ w.p. p_{neg} , else $\mathcal{S}_+$; $S_t \sim \text{Uniform}(\mathcal{J})$; $k \leftarrow \min(K - 1, T - t)$ {first forced step; K steps total, sign per step}
- 13: **end if**
- 14: **end if**
- 15: Draw $G_t \sim \mu_{S_t} + \sigma_{S_t} t_5$ {Eq. (2)}
- 16: **end for**

only paths with at least one jump (Glasserman, 2003). The kurtosis weight w_κ balanced the kurtosis and ACF terms. The search ranges, w_κ , L , and the remaining HMM-WJ and SIM hyperparameters are reported in Table 1. The fitted model was evaluated against the marginal and temporal statistics reported in Section 4.

3.2. Variance-corrected multi-asset composition

The single-asset model describes each ticker separately. It does not say how several tickers should move together. For the multi-asset extension, we added one common source of movement: a market path. We used the single-index model (SIM) of Sharpe (1963):

$$g_i(t) = \alpha_i + \beta_i g_m(t) + \varepsilon_i(t), \quad t = 1, \dots, T. \quad (4)$$

Here α_i is the intercept for asset i , g_m is the market growth-rate path, β_i is the asset's market loading, and ε_i is its asset-specific residual path. The symbol ε_i denotes a residual path; it is unrelated to the jump probability ϵ used by the single-asset model. We estimated the intercept and market loading (α_i, β_i) from real data by ordinary least squares (OLS). The remaining modeling choice is ε_i . A standard SIM draws independent Gaussian residuals with the variance estimated by the real-data regression. This preserves the fitted market relationship but discards the heavy tails and volatility regimes produced by the single-asset generator. We called this baseline *Gaussian SIM*.

Why direct construction fails.. A tempting alternative is to use a full synthetic return path $\tilde{g}_i$ from the single-asset generator as the residual:

$$g_i = \alpha_i + \beta_i g_m + \tilde{g}_i. \quad (5)$$

This raw substitution double counts location as well as variance. The calibrated intercept already fixes the conditional location of asset i , whereas the full-return draw generally has $\mathbb{E}[\tilde{g}_i] \neq 0$. Equation (5) therefore has conditional mean $\alpha_i + \mathbb{E}[\tilde{g}_i] + \beta_i g_m$, not $\alpha_i + \beta_i g_m$.

All full-return constructions in our comparison first removed the realized path mean. We defined the centered generator path as follows:

$$\bar{\tilde{g}}_i \equiv \frac{1}{T} \sum_{t=1}^T \tilde{g}_i(t), \quad \tilde{g}_i^c(t) \equiv \tilde{g}_i(t) - \bar{\tilde{g}}_i. \quad (6)$$

Therefore, $T^{-1} \sum_t \tilde{g}_i^c(t) = 0$ exactly. Subtracting a constant leaves both the sample variance and the sample covariance with g_m unchanged. We therefore used the centered direct construction $g_i = \alpha_i + \beta_i g_m + \tilde{g}_i^c$ as the *naive* variance comparator. Centering isolates the variance-composition question: the naive method has the correct residual location but no variance correction.

The path variance $\sigma_{\text{gen},i}^2 = \text{Var}(\tilde{g}_i) = \text{Var}(\tilde{g}_i^c)$ is the *generator variance*. It already represents the asset's total variance. Adding the market term therefore contributes variance a second time. Because $\tilde{g}_i$ was sampled independently of g_m , the naive composed variance was as follows:

$$\text{Var}(g_i) = \beta_i^2 \sigma_m^2 + \sigma_{\text{gen},i}^2, \quad \text{Cov}(\tilde{g}_i^c, g_m) \approx 0, \quad (7)$$

where $\sigma_m^2 = \text{Var}(g_m)$ is the variance of the market path. The excess is $\beta_i^2 \sigma_m^2$, so the error is largest for assets with high market exposure.

Variance-preserving correction.. We treated the generator variance for asset i , $\sigma_{\text{gen},i}^2$, as the total variance budget for the composed asset. The market term uses part of that budget. We scaled the single-asset draw so that it uses only the part that remains. We defined s_i as the asset-specific residual scale factor, set $\varepsilon_i = s_i \tilde{g}_i^c$, and required $\text{Var}(\beta_i g_m + s_i \tilde{g}_i^c) = \sigma_{\text{gen},i}^2$. Solving for s_i gives the correction:

$$s_i^2 = 1 - \frac{\beta_i^2 \sigma_m^2}{\sigma_{\text{gen},i}^2} \quad (8)$$

We defined the *market loading ratio* as follows:

$$\rho_i \equiv \frac{\beta_i^2 \sigma_m^2}{\sigma_{\text{gen},i}^2}, \quad (9)$$

This ratio is the fraction of the variance budget used by the market. The residual receives the remaining fraction, so $s_i^2 = 1 - \rho_i$. This correction is impossible when $\rho_i \geq 1$ because the market term alone uses the entire variance budget.

This case can occur when the market loading β_i is large, the per-asset generator has low variance, or the synthetic market is more volatile than the calibration market. In that case we reserved a minimum fraction f , the idiosyncratic variance floor, for the asset-specific draw and reduced the effective market loading β_i^{eff} . Equivalently, the maximum allowable market loading ratio is $\rho_i^{\text{max}} = 1 - f$. The resulting rule was:

$$\begin{aligned}\beta_i^{\text{eff}} &= \begin{cases} \beta_i & \text{if } \rho_i \leq \rho_i^{\text{max}}, \\ \text{sign}(\beta_i) \sqrt{(1-f) \frac{\sigma_{\text{gen},i}^2}{\sigma_m^2}} & \text{if } \rho_i > \rho_i^{\text{max}}, \end{cases} \\ s_i^2 &= \begin{cases} 1 - \rho_i & \text{if } \rho_i \leq \rho_i^{\text{max}}, \\ f & \text{if } \rho_i > \rho_i^{\text{max}}. \end{cases}\end{aligned}\tag{10}$$

The sign of the calibrated market loading β_i is retained. We used $f = 0.10$, which guarantees that at least 10% of the variance budget remains asset-specific (Table 1).

High- R^2 index trackers. For an ordinary stock, our main target was total variance. For an index-tracking fund, preserving its relationship with the market was more important. The coefficient of determination R^2 measures the fraction of the asset's variance explained by the market. For example, SPY regressed on itself has $\beta = 1$, $R^2 = 1$, and no residual variance. The QQQ and SPYG funds also have high R^2 against SPY. Applying the ordinary-stock rule to these trackers can preserve total variance while changing their market relationship. We therefore used a separate branch when the calibrated coefficient of determination, $R_{i,\text{real}}^2$, is at least the preservation threshold R_{preserve}^2 . In our universe, 0.80 separated the index trackers from individual stocks (Supplementary Fig. S3). On this branch, we computed the target residual variance needed to recover the calibrated R^2 as follows:

$$\sigma_{\varepsilon,\text{target}}^2 = \beta_i^2 \sigma_m^2 \cdot \frac{1 - R_{i,\text{real}}^2}{R_{i,\text{real}}^2}\tag{11}$$

We rescaled the generator draw to this target as follows:

$$\varepsilon_i = \left(\sqrt{\frac{\sigma_{\varepsilon,\text{target}}^2}{\sigma_{\text{gen},i}^2}} \right) \tilde{g}_i^c.\tag{12}$$

When $R_{i,\text{real}}^2 = 1$, the target residual variance is zero, as it should be for SPY against itself. If the scale factor in Eq. (12) exceeds one, the procedure stretches the generator draw. We flagged this case because it can weaken the original marginal fit.

Algorithm 2 Hybrid SIM composition with variance correction.

Require: Market growth-rate path $g_m \in \mathbb{R}^T$ with sample variance σ_m^2 ; per-asset generator growth-rate paths $\tilde{g}_i \in \mathbb{R}^T$ for $i = 1, \dots, N$; calibrated SIM parameters $(\alpha_i, \beta_i, R_{i,\text{real}}^2)$; idiosyncratic variance floor $f \in (0, 1)$; R^2 -preservation threshold $R_{\text{preserve}}^2 \in (0, 1)$.

Ensure: Composed paths $g_i \in \mathbb{R}^T$ and per-asset construction flags.

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1: for each asset  $i = 1, \dots, N$  do
2:   Compute generator variance  $\sigma_{\text{gen},i}^2 \leftarrow \text{Var}(\tilde{g}_i)$ .
3:   Center the full-return draw:  $\tilde{g}_i^c \leftarrow \tilde{g}_i - T^{-1} \sum_{t=1}^T \tilde{g}_i(t)$ .
4:   if  $R_{i,\text{real}}^2 \geq R_{\text{preserve}}^2$  then
5:      $\sigma_{\varepsilon,\text{target}}^2 \leftarrow \beta_i^2 \sigma_m^2 (1 - R_{i,\text{real}}^2) / R_{i,\text{real}}^2$  {Eq. (11)}
6:      $\varepsilon_i \leftarrow \left( \sqrt{\sigma_{\varepsilon,\text{target}}^2 / \sigma_{\text{gen},i}^2} \right) \tilde{g}_i^c$ 
7:      $\beta_i^{\text{eff}} \leftarrow \beta_i$ ; branch  $\leftarrow R^2$ -preserving
8:   else
9:      $\rho_i \leftarrow \beta_i^2 \sigma_m^2 / \sigma_{\text{gen},i}^2$  {Eq. (9)}
10:    if  $\rho_i \leq 1 - f$  then
11:       $s_i^2 \leftarrow 1 - \rho_i$ ;  $\beta_i^{\text{eff}} \leftarrow \beta_i$ ; branch  $\leftarrow$  variance-preserving
12:    else
13:       $s_i^2 \leftarrow f$ ;  $\beta_i^{\text{eff}} \leftarrow \text{sign}(\beta_i) \sqrt{(1 - f) \sigma_{\text{gen},i}^2 / \sigma_m^2}$ ; branch  $\leftarrow$  clipped
        variance-preserving
14:    end if
15:     $\varepsilon_i \leftarrow s_i \tilde{g}_i^c$ 
16:  end if
17:  (Optional) apply cross-sectional rank reorder to  $\varepsilon_i$ .
18:   $g_i \leftarrow \alpha_i + \beta_i^{\text{eff}} g_m + \varepsilon_i$  {Eq. (4)}
19: end for

```

Complete composition rule.. Algorithm 2 puts these steps together. For a high- R^2 tracker, it preserves the calibrated R^2 . For other assets, it preserves the generator variance and clips beta only when the market term would leave too little asset-specific variance. It then adds the scaled asset draw to the market term. An optional rank reorder can add dependence among the residual draws without changing their individual variances. Centering kept the path mean controlled by the intercept and market term. Under the zero-covariance assumption, an OLS regression on a long composed path therefore recovered the effective market loading and its corresponding intercept. The selected branch also fixed the target variance. These are guarantees about variance and factor loading, not about the complete return distribution. Online Appendix S8 gives the algebra. The construction adds only same-day market dependence; it does not reproduce lead-lag effects or market-driven volatility clustering.

Table 1: **HMM-WJ and SIM composition hyperparameters.** Values used for the main SPY analysis, with the sensitivity-sweep or grid-search span and the location of the corresponding check where one exists. The 423-ticker SIM composition universe holds $\epsilon = 0$ instead of the SPY value; per-ticker (ϵ^*, λ^*) for the NVDA, JNJ, and JPM cross-asset validation are given in Supplementary Table S7.

Symbol	Description	SPY value	Explored range
<i>State partition and emission</i>			
N	Laplace-quantile states	100	$\{30, \dots, 350\}$, Online Appendix S6
ν	Student- t emission degrees of freedom	5	$\{3, \dots, 30, \infty\}$
<i>Jump-duration override</i>			
ϵ	jump probability per step	10^{-4}	$[10^{-4}, 2.5 \times 10^{-2}]$, Eq. (3)
λ	mean jump-episode duration (steps)	90	$[10, 160]$, Eq. (3)
p_{neg}	probability a jump targets $\mathcal{S}_-$	0.52	fixed
N_{tail}	quantile states per tail set	5	fixed (at $N = 100$)
w_κ	kurtosis-penalty weight, Eq. (3)	0.20	fixed
L	ACF lag window, Eq. (3)	25 days	fixed
<i>SIM composition</i>			
f	idiosyncratic variance floor	0.10	$\{0.05, \dots, 0.30\}$
R_{preserve}^2	branch-selection threshold	0.80	$\{0.70, \dots, 0.90\}$

Evaluation protocol. We evaluated the method on 424 US equities and exchange-traded funds from Polygon.io. We selected tickers with a complete sequence of 2,767 daily closing prices from January 2014 through December 2024, which yielded 2,766 growth rates. This complete-case selection conditions the analysis on assets that survived and retained data coverage throughout the training window. We used SPY as the market and treated the other 423 tickers as assets to compose. For each asset, we fitted the same state and emission model used in the SPY analysis (Table 1; $r_f = 0$). To focus the experiment on multi-asset composition rather than per-ticker jump calibration, we set the jump probability to zero. We also regressed each real asset series on SPY to estimate the intercept α_i , market loading β_i , coefficient of determination $R_{i,\text{real}}^2$, and residual variance. Across the universe, β ranged from 0.03 to 1.79 and $R_{i,\text{real}}^2$ ranged from 0.001 to 0.933 (Supplementary Fig. S3).

We compared six ways to construct the asset-specific term. The naive method used the full JumpHMM draw without correction. The Gaussian SIM method used independent Gaussian residuals with variance estimated by OLS. The hybrid method used Algorithm 2. The JumpHMM-on-residuals method fitted a new JumpHMM to the OLS residual series after converting the residuals to a synthetic price path by cumulative compounding. The block-bootstrap method sampled the OLS residual series with the Politis–Romano stationary bootstrap (Politis and Romano, 1994) and a mean block length of $\sqrt{T} \approx 50$. The GARCH(1,1)- t method fitted a conditional-variance model to each residual series (Bollerslev, 1986). We excluded thirty non-stationary fits from the GARCH results only. For each ticker, the naive and hybrid methods received the same JumpHMM draw. This pairing isolated the effect of the variance correction. The other four methods generated their own residual draws. We did not use the optional cross-sectional rank reorder because the experiment tested each ticker’s marginal

construction separately.

We asked three questions about each composed series. First, did a new OLS regression recover the calibrated market loading and R^2 ? We measured this with the absolute errors $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R_{\text{real}}^2|$, where lower values are better. Second, did the synthetic and real returns have similar distributions? We used Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) pass rates, Wasserstein-1 distance, excess kurtosis, and the Hill upper-tail index. Higher KS and AD pass rates and lower Wasserstein distance indicate a closer match; kurtosis and the Hill index describe tail behavior. Third, did the composed path preserve generator variance? For this question, a composed-to-generator variance ratio of one is ideal. Online Appendix S3 gives the metric definitions. We ran 100 replications per ticker with seed 1234, producing 42,300 series per method. Pass rates used the significance level $\alpha = 0.05$.

We then froze all marginal fits, residual models, and SIM parameters estimated from 2014–2024 and evaluated the methods against the 249 growth rates observed in 2025. Of the 423 non-market training assets, 416 had complete holdout histories; GARCH results covered 386 because its thirty non-stationary training fits remained excluded. For each replication, we generated a 249-day SPY path from the training-period SPY marginal and used that common simulated market path in all asset compositions. Observed 2025 SPY and asset returns were used only for scoring. We repeated the KS, AD, Wasserstein-1, variance, kurtosis, and one-day left-tail VaR evaluations with 100 paths per ticker. Because goodness-of-fit tests have less power on 249 observations than on 2,766, we also compared each synthetic path with a randomly selected contiguous 249-day block from the training history. We first averaged replication-level results within ticker and then summarized across tickers so each asset received equal weight.

4. Results

4.1. Single-asset generator validation

We first tested whether the marginal generator in Fig. 2 reproduced the empirical SPY targets shown in Fig. 1. In both the in-sample (2014–2024, $T = 2,766$) and out-of-sample (2025, $T = 249$) windows, the real SPY series had heavy tails, and the Jarque-Bera test rejected normality. The series also showed persistent volatility clustering. Individual raw-return correlations were small, although the Ljung–Box test detected joint serial dependence (Supplementary Table S1). We calibrated (ϵ, λ) for HMM-WJ against the in-sample SPY series using the grid search in Eq. (3) (Algorithm 1; search range in Table 1). The selected values were $(\epsilon, \lambda) = (10^{-4}, 90)$, although nearby grid points gave similar objective values. Under the counted transition matrix, ordinary state residence times were only 1–2 steps, whereas the selected jump duration had a mean of 90 steps (Supplementary Fig. S5). The primary analysis used $N = 100$ states. Across $N = 60$ to 200, the KS and AD pass rates varied by less than one percentage point for both HMM-NJ and HMM-WJ. At $N = 350$, some states

Table 2: **In-sample and out-of-sample model comparison for SPY.** Results summarize 1,000 simulated paths. Values in parentheses are standard errors, and bold entries are best among the fitted generative models within each panel; the empirical bootstrap is excluded from this highlighting. Metric definitions and standard-error calculations are given in Online Appendix S3.

Model	KS (%)	AD (%)	κ	ACF-MAE	Cov. (%)	W_1	Hellinger
<i>In sample: 2014–2024 (2,766 days; observed $\kappa = 7.715$)</i>							
Bootstrap	100.0 (<0.1)	99.7 (0.2)	7.6 (0.08)	0.060 (<0.001)	100.0 (<0.1)	0.062 (0.001)	0.042 (<0.001)
Gaussian	0.0 (<0.1)	0.0 (<0.1)	−0.0 (<0.01)	0.060 (<0.001)	13.1 (0.1)	0.399 (0.001)	0.148 (<0.001)
Laplace	44.0 (1.6)	43.3 (1.6)	3.0 (0.02)	0.060 (<0.001)	37.4 (1.5)	0.138 (0.001)	0.072 (<0.001)
GARCH(1,1)	5.5 (0.7)	1.9 (0.4)	8.2 (0.37)	0.031 (0.001)	29.3 (1.4)	0.304 (0.006)	0.100 (0.001)
GRU	0.6 (0.2)	0.2 (0.1)	5.7 (0.16)	0.036 (<0.001)	17.2 (0.5)	0.421 (0.003)	0.134 (<0.001)
HSMM	82.0 (1.2)	42.5 (1.6)	4.8 (0.08)	0.059 (<0.001)	68.7 (1.1)	0.176 (0.001)	0.113 (<0.001)
HMM-NJ	99.3 (0.3)	98.2 (0.4)	8.1 (0.14)	0.059 (<0.001)	100.0 (<0.1)	0.081 (0.001)	0.072 (<0.001)
HMM-WJ	98.3 (0.4)	94.8 (0.7)	7.5 (0.09)	0.053 (<0.001)	100.0 (<0.1)	0.097 (0.001)	0.074 (<0.001)
<i>Out of sample: 2025 (249 days; observed $\kappa = 6.867$)</i>							
Bootstrap	97.4 (0.5)	98.5 (0.4)	6.0 (0.18)	0.043 (<0.001)	100.0 (<0.1)	0.232 (0.002)	0.205 (0.001)
Gaussian	62.2 (1.5)	46.3 (1.6)	−0.0 (<0.01)	0.043 (<0.001)	36.4 (1.9)	0.452 (0.002)	0.235 (0.001)
Laplace	88.0 (1.0)	92.9 (0.8)	2.7 (0.05)	0.043 (<0.001)	74.7 (1.0)	0.263 (0.002)	0.211 (0.001)
GARCH(1,1)	80.3 (1.3)	72.0 (1.4)	1.6 (0.07)	0.026 (<0.001)	96.0 (1.3)	0.507 (0.018)	0.232 (0.001)
GRU	71.8 (1.4)	44.8 (1.6)	2.9 (0.10)	0.033 (<0.001)	77.8 (2.9)	0.488 (0.005)	0.240 (0.001)
HSMM	96.2 (0.6)	96.7 (0.6)	4.1 (0.13)	0.042 (<0.001)	100.0 (0.2)	0.287 (0.002)	0.239 (0.001)
HMM-NJ	96.6 (0.6)	96.1 (0.6)	6.2 (0.15)	0.041 (<0.001)	100.0 (<0.1)	0.259 (0.003)	0.207 (0.001)
HMM-WJ	95.4 (0.7)	95.3 (0.7)	6.2 (0.15)	0.040 (<0.001)	100.0 (<0.1)	0.275 (0.005)	0.210 (0.001)

KS and AD report pass rates at $\alpha = 0.05$; coverage is the fraction of empirical quantiles inside the synthetic 90% envelope. κ is simulated excess kurtosis. ACF-MAE is computed for $|G_t|$ through lag 252. Larger values are preferred for KS, AD, and coverage; smaller values are preferred for ACF-MAE, W_1 , and Hellinger distance; κ is compared with the observed value.

were never visited and their emission parameters could not be estimated (Online Appendix S6). The results were stable across a broad range of state resolutions, so we kept $N = 100$ for the generator comparisons that followed.

With the model calibrated, we compared HMM-WJ with seven alternative generators on 1,000 paths of 2,766 trading days each (Table 2; Fig. 3). No baseline was best on both the return distribution and volatility clustering. Bootstrap resampling passed every distribution test but, like all independent sampling methods, did not reproduce clustering. Gaussian sampling passed none of the distribution tests. The GARCH(1,1) model had the lowest mean absolute autocorrelation error (ACF-MAE) of 0.031 but rarely passed the distribution tests. The gated recurrent unit (GRU) (Cho et al., 2014) showed the same general pattern. The HSMM-style baseline adapted from Bulla and Bulla (2006) reached an 82% KS pass rate but underestimated kurtosis by more than one third.

Among the fitted parametric and latent-state generators, the two HMM variants gave the strongest overall distributional fit. The HMM-NJ and HMM-WJ models had KS pass rates of 99.3% and 98.3%, respectively. The slightly lower HMM-WJ pass rates were consistent with jump episodes sometimes placing too much probability in the tails. Student- t emissions produced simulated kurtosis of 7.5, close to the observed 7.7, whereas Normal emissions underestimated it by about 29% (Supplementary Table S3; Online Appendix S5). Both HMM variants also reproduced the observed skewness of -0.75 . That skewness arose

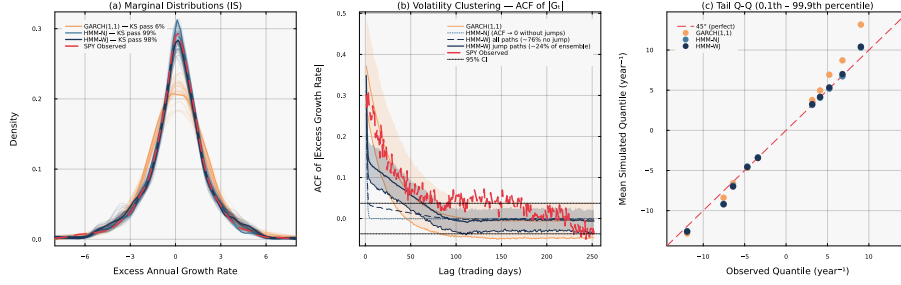


Figure 3: **Head-to-head in-sample model comparison for SPY** ($N = 100$, 1,000 simulated paths). *Panel (a)*: marginal density of excess growth rates with in-sample KS pass rates annotated per generator. *Panel (b)*: autocorrelation function of $|G_t|$ at lags 1–252 (shaded bands: 10th–90th percentile across paths) per generator. *Panel (c)*: tail Q-Q plot at the 0.1st–99.9th percentile region per generator.

from unequal time spent in negative and positive states rather than from an explicit skewness parameter.

The jump episodes improved temporal fit. The HMM-WJ model reduced ACF-MAE from 0.059 for HMM-NJ to 0.053. About 24% of HMM-WJ paths contained at least one jump and showed more persistent autocorrelation in absolute returns; paths without a jump were equivalent to HMM-NJ. On the held-out 249 days of 2025, HMM-NJ and HMM-WJ retained KS pass rates of 96.6% and 95.4%, with kurtosis close to the observed value (Table 2; Supplementary Fig. S4). Tests on NVDA, JNJ, and JPM showed strong in-sample fit but more variable holdout performance (Supplementary Table S7, Online Appendix S7). These comparisons support HMM-WJ as a compromise between distributional and temporal fit. We examined that compromise directly by varying the share of jump-active paths.

To measure how jump-active paths changed the results, we resampled the fitted ensemble at different jump-path fractions f (Fig. 4). Here f is the share of accepted paths that contain at least one jump. We varied f from 0, where all accepted paths reduce to HMM-NJ, to 1, where every accepted path contains a jump. This procedure held the fitted model fixed and changed only the mixture of accepted paths. The absolute-return autocorrelation error fell monotonically from 0.059 at $f = 0$ to 0.037 at $f = 1$, moving from the no-jump value toward the GARCH value as the jump share rose. Over the same sweep, simulated excess kurtosis declined from 7.9 to 6.7, matching the observed value near $f \approx 0.14$ and falling below it thereafter. The two-sample KS and AD pass rates also declined. The Hill upper-tail index stayed near 0.35, against an observed 0.32, across the whole sweep because the Student- t ($\nu = 5$) emission sets the tail exponent independently of jump content. Increasing the jump share therefore traded fit to the distributional body for lower autocorrelation error without changing the tail index.

The value $f \approx 0.25$ used throughout kept simulated kurtosis within 2% of the observed value and lowered the autocorrelation error about a quarter of the

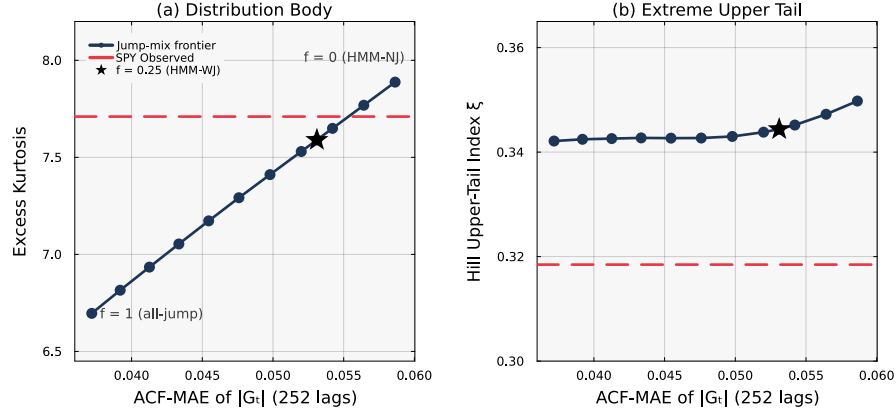


Figure 4: **Jump-mix enrichment frontier for SPY** (in-sample, $N = 100$, pool of 4,000 simulated paths). Resampling the fitted HMM-WJ ensemble to a target jump-path fraction f (the share of accepted paths containing at least one jump) traces the temporal-marginal tradeoff. Both panels share the horizontal axis: absolute-return autocorrelation error (ACF-MAE of $|G_t|$ over lags 1–252); lower values indicate stronger volatility clustering. Panel (a): simulated excess kurtosis. Panel (b): Hill upper-tail index ξ , the raw estimator equal to the reciprocal of the tail exponent. Red dashed lines mark the observed SPY values; the star marks the $f \approx 0.25$ value used throughout. Endpoints $f = 0$ and $f = 1$ correspond to the no-jump generator and an all-jump ensemble.

way from the no-jump value toward the all-jump endpoint. The selected mixture therefore provided a compromise between marginal and temporal fit before the fitted marginals were carried into the multi-asset composition experiment.

4.2. Variance-corrected multi-asset composition

The single-asset results defined the tradeoff in the marginal generator. We then tested whether the variance-corrected SIM preserved each asset’s variance and factor loading across the 423-ticker universe, comparing the six composition methods on the aggregate scorecard (Table 3). Before comparing the methods, we measured the cross-term omitted from the variance correction. For the full-return generator, the universe-mean normalized cross-covariance with the market path was 1.53×10^{-4} , and the per-ticker values ranged from -0.0061 to 0.0061 (Supplementary Fig. S1). These values supported the negligible-cross-term approximation used in Eq. (8).

Table 3: **Aggregate scorecard across 423 non-market tickers and 100 replications per ticker.** Errors, distances, and tail metrics are medians per method. Pass fractions for KS and AD are pooled across replications. $|\hat{\alpha} - \alpha|$, $|\hat{\beta} - \beta|$, and $|\hat{R}^2 - R_{\text{real}}^2|$: absolute deviation between OLS recovery and calibration (intercept error has units of yr^{-1}). KS pass and AD pass: fraction of replications whose two-sample test against the real marginal exceeds $p = 0.05$. W_1 : Wasserstein-1 distance to the real marginal. κ : sample excess kurtosis. Hill: tail-index estimate on the upper 5% of $|g|$.

Method	$ \hat{\alpha} - \alpha $	$ \hat{\beta} - \beta $	$ \hat{R}^2 - R_{\text{real}}^2 $	KS pass (%)	AD pass (%)	W_1	κ	Hill
Naive	0.002	0.022	0.087	7.1	3.5	0.700	7.767	0.369
Gaussian SIM	0.048	0.017	0.008	0.6	0.5	0.682	1.427	0.217
Hybrid	0.002	0.018	0.021	73.6	64.6	0.249	7.165	0.365
JumpHMM-on-residuals	0.049	0.018	0.015	75.8	70.7	0.232	7.262	0.365
Block bootstrap	0.042	0.017	0.020	73.3	66.8	0.232	6.675	0.330
GARCH(1,1)- t	0.047	0.017	0.036	67.4	61.4	0.267	6.069	0.317

All six methods recovered the factor loading with similar accuracy: the median absolute β error ranged from 0.017 to 0.022 (Table 3). Path centering also kept the median absolute intercept error at 0.002 yr^{-1} for both methods that used full-return draws. The residual-fit methods retained random variation in the residual mean over each finite path. Their larger intercept errors therefore do not, by themselves, imply a worse fit to the return distribution.

The larger differences appeared in variance recovery and tail behavior. Gaussian SIM had the lowest median R^2 error because it targets R^2 directly. However, it almost never passed the distributional tests, and its median excess kurtosis was an order of magnitude below the observed value (Supplementary Fig. S2). Centered naive composition retained the shape of the full-return draw, but adding the market factor inflated its variance by $\beta_i^2 \sigma_m^2$. The hybrid method removed that extra variance. It passed 73.6% of KS comparisons and retained heavy tails, with median excess kurtosis of 7.2 and a Hill index of 0.36. Its median R^2 error fell between those of the naive and Gaussian methods because this branch preserves the generator variance rather than targeting the observed R^2 directly.

The three residual-fit methods avoided double counting by fitting their generators to the empirical OLS residuals, whose variance already excludes the market component. Their KS pass rates ranged from 67.4% for GARCH(1,1)- t to 75.8% for JumpHMM-on-residuals. The hybrid rate was two percentage points below JumpHMM-on-residuals and slightly above block bootstrap. JumpHMM-on-residuals and block bootstrap had slightly lower Wasserstein distances. JumpHMM-on-residuals also matched the hybrid’s kurtosis and Hill index almost exactly, whereas the block-bootstrap and GARCH residual fits produced lighter tails. The correction substantially improved on naive composition and approached the best residual-fit method without requiring a second fit for each asset. We next tested whether this ordering persisted after all fitted parameters were frozen.

We froze the fitted models and SIM parameters before evaluating them against 416 complete asset histories from 2025 (Table 4). The hybrid method reached an 87.0% mean cross-ticker KS pass rate, against 83.5% for centered naive composition, and its median composed-to-observed variance ratio was 0.97,

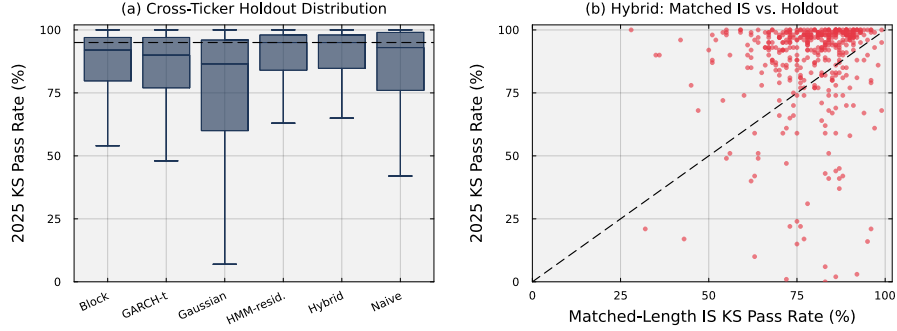


Figure 5: **Cross-ticker distribution of frozen-parameter out-of-sample marginal fit.** Panel (a) shows the distribution of per-ticker 2025 KS pass rates across 100 replications; the dashed line marks 95%. Panel (b) compares each ticker’s hybrid KS pass rate against a matched-length training block with its pass rate against the 2025 holdout. The diagonal denotes equal performance.

whereas naive composition remained inflated at 1.29. JumpHMM-on-residuals reached an 86.2% KS pass rate and a slightly lower median W_1 than the hybrid. Matched-length training blocks produced lower pass rates for every method. The higher 2025 pass rates did not arise solely from comparing samples of 249 rather than 2,766 observations.

The average pass rates concealed substantial differences among tickers (Fig. 5). Most ticker-level rates were near the upper end of the range, but every method retained a lower tail of weak fits. For the hybrid method, many tickers improved relative to their matched-length training blocks, while a subset deteriorated out of sample. The held-out marginal improvement was broad but not uniform across assets, motivating a separate check of whether the resulting synthetic tails produced reliable risk thresholds.

Table 4: **Frozen-parameter multi-asset evaluation on the 2025 holdout.** Results cover 416 non-market tickers and 100 replications per ticker; GARCH covers 386 tickers. Pass rates are first averaged within ticker and then across tickers. *matched IS* compares each synthetic path with a randomly selected contiguous 249-day block from the 2014–2024 training history, matching the holdout sample length. W_1 and the variance ratio are medians across ticker-level medians; the variance ratio is synthetic variance divided by observed 2025 variance.

Method	KS pass (%)		AD pass (%)		W_1 (2025)	variance ratio
	matched IS	2025	matched IS	2025		
Naive	69.4	83.5	54.8	71.6	0.877	1.29
Gaussian SIM	57.4	73.9	44.9	64.6	0.866	0.97
Hybrid	78.8	87.0	66.2	79.7	0.716	0.97
JumpHMM-on-residuals	77.9	86.2	65.6	80.1	0.708	0.98
Block bootstrap	76.3	84.0	64.0	76.5	0.715	0.90
GARCH(1,1)-t	74.5	82.6	62.6	75.0	0.733	0.91

4.3. VaR coverage

We used the frozen synthetic paths to estimate per-ticker one-day left-tail Value-at-Risk (VaR) thresholds and counted exceedances on the real 2025 histories (Table 5). At $\alpha = 0.95$, the hybrid and JumpHMM-on-residuals methods produced mean exceedance rates of 5.97% and 6.07%, respectively, against the 5% nominal rate. At $\alpha = 0.99$, both rates were 1.67%, against the 1% nominal rate. Naive composition was closest at the 99% level (1.21%), while Gaussian SIM had the largest exceedance rate (2.03%). The short holdout contains only about 2–3 expected 99% exceedances per ticker, so we interpret the corresponding Kupiec pass rates and cross-ticker dispersion as low-power checks rather than precise rankings. Taken together, the VaR results showed that stronger marginal fit did not automatically imply nominal tail coverage, motivating a closer look at where the variance correction was most stable.

Table 5: **Out-of-sample per-ticker one-day left-tail Value-at-Risk coverage across the six composition methods.** VaR thresholds at coverage level α were estimated from frozen-parameter synthetic paths and exceedances were counted against the real 2025 growth-rate history. *rate*: mean of the per-ticker mean exceedance rates (nominal: $1 - \alpha$). *SD*: cross-ticker standard deviation, in percentage points. *Kupiec pass*: mean within-ticker fraction of replications whose unconditional-coverage test p -value exceeds 0.05.

Method	$\alpha = 0.95$			$\alpha = 0.99$		
	rate (%)	SD (pp)	Kupiec pass (%)	rate (%)	SD (pp)	Kupiec pass (%)
Naive	4.37	2.15	65.8	1.21	0.76	78.9
Gaussian SIM	4.95	2.26	70.0	2.03	1.17	72.1
Hybrid	5.97	2.43	66.3	1.67	0.93	73.4
JumpHMM-on-residuals	6.07	2.59	67.0	1.67	0.95	74.5
Block bootstrap	6.13	2.56	63.7	1.84	1.01	73.4
GARCH(1,1)- t	5.79	2.47	62.4	1.80	1.02	72.0

4.4. Sensitivity analyses

To isolate how the variance correction changed marginal fit, we plotted each ticker’s KS pass rate and variance ratio $\text{Var}(g)/\sigma_{\text{gen}}^2$ against its calibrated β (Fig. 6). Under naive composition, the variance ratio followed $1 + \rho_i$ and exceeded generator variance by 50–75% at high β , and the naive KS pass rate fell toward zero above $\beta \approx 0.7$; the hybrid method held the ratio at one and retained substantially higher pass rates across the β range.

The branch-selection rule assigned 421 tickers to the variance-preserving branch, two trackers to the R^2 -preserving branch, and none to the clipped branch (Supplementary Figs. S3 and S6; Supplementary Table S4). The two R^2 -preserving assignments were QQQ and SPYG, and both had a 100% KS pass rate; the highest- R^2 individual stock, BLK, fell about 0.16 below the $R^2_{\text{preserve}} = 0.80$ threshold. Clipping did not occur in the universe: $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\text{gen},i}^2$ (Eq. (9)) stayed below $1 - f = 0.90$ for every ticker. Because the R^2 -preserving branch contained only two tickers, we tested it on a synthetic-tracker grid spanning $(\beta_{\text{true}}, R^2_{\text{true}}) \in \{0.8, 1.0, 1.2\} \times \{0.80, 0.85, 0.90, 0.95, 0.99\}$. Each cell had a 100% KS pass rate across replications. The correction removed the observed beta-related variance inflation. We then tested the clipping branch under conditions that would force it to activate.

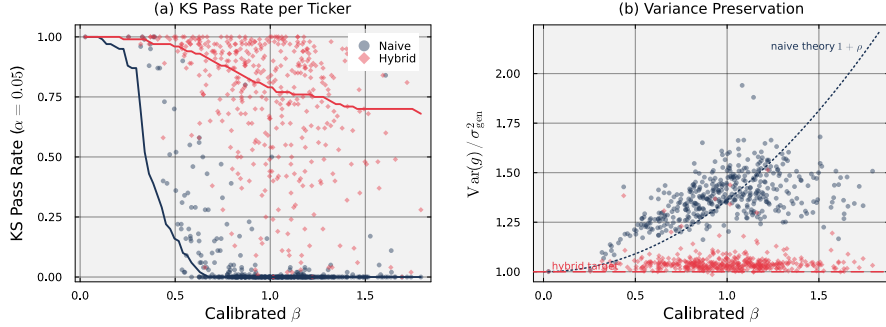


Figure 6: **Hybrid’s variance correction and its KS-pass-rate consequence, per ticker vs calibrated β .** Each point is a per-ticker median over 100 replications; thick lines are binned medians. *Left (consequence):* Kolmogorov–Smirnov pass rate against the real marginal at $\alpha = 0.05$ for naive composition (blue) and hybrid (orange). *Right (cause):* variance ratio $\text{Var}(g)/\sigma_{\text{gen}}^2$ for the same two methods. The dotted curve is the theoretical naive inflation $1 + \rho$ evaluated at the universe-median σ_{gen}^2 ; the dashed line marks a variance ratio of one.

Bucketing the universe by β quartile, the hybrid KS pass rate fell from 87.6% in the lowest quartile to 65.9% in the highest (Supplementary Table S5). As β increased, the market term accounted for more of the composed variance, making the marginal more dependent on the market distribution. We tested the clipping rule directly by rescaling the market growth-rate path by $\gamma \in \{1, 2, 3\}$, holding the calibrated marginals fixed (Supplementary Table S6). At $\gamma = 1$ no ticker crossed the threshold. At $\gamma = 2$ and 3, clipping applied to 76.4% and 95.2% of tickers. The clipping rule was unnecessary under observed market variance but activated as intended when market volatility was artificially increased. At the observed market variance, the floor f did not affect any branch assignment; varying R_{preserve}^2 changed only the two tracker assignments identified above.

4.5. Daily-rebalanced portfolio check

To test a use that depends on joint cross-asset behavior, we applied a 20-asset target-weight allocator to the hybrid-composed 2025 paths. Switching the same target weights from static allocation to daily rebalancing increased the median annualized growth rate from 1.97% to 29.69%. Adding the allocation signal increased it by only another 0.13 percentage points, to 29.83%. The same allocator on the real 2025 history returned 11.50% (Supplementary Table S2). This comparison was consistent with omitted residual covariance creating a diversification-return bias, but it did not isolate how much of the gap that omission caused. A constant log-wealth shift set the synthetic ensemble median to the prespecified 8% annual return prior, but it did not restore the missing covariance.

5. Discussion

The single-asset analysis supplied the return generator used in the multi-asset construction. Relative to HMM-NJ, HMM-WJ reduced the autocorrelation error for absolute returns, although its in-sample KS and AD pass rates were slightly lower. The GARCH model achieved the lowest autocorrelation error of any generator but rarely passed the distributional tests, and the GRU showed a similar tradeoff. The adapted HSMM-style baseline passed the KS test more often but produced too little kurtosis. Bootstrap resampling passed the distributional tests by drawing directly from the observed returns, but it did not reproduce volatility clustering. We therefore interpret HMM-WJ as a compromise between distributional and temporal fit, not as a uniformly better single-asset model.

The distributional pass rates also need cautious interpretation. The two-sample KS and AD tests assume independent observations, whereas the HMM, HSMM, GARCH, and block-bootstrap paths contain dependence. For that reason, we considered the tests alongside Wasserstein and Hellinger distances and the Hill tail index. Model resolution creates a second qualification. A sensitivity sweep over $N \in \{30, 60, 90, 100, 150, 200\}$ showed little change across the middle of this range for the 2,766-day SPY sample, but shorter histories will support fewer well-populated states. We also tuned the jump parameters (ϵ, λ) on SPY and then applied them across the asset universe. Separate fits for NVDA, JNJ, and JPM achieved in-sample KS pass rates of at least 98.9%, but three assets cannot establish that one jump calibration transfers to every return series. Finally, the Student- t emissions model heavy tails but not skewness directly. The fitted skewness instead came from unequal time spent in negative and positive states, so assets with different asymmetry may need a different state partition.

The main multi-asset result was the variance correction. Naive composition added market variance to a draw that already represented the asset’s total variance. The correction removed that double count and passed 73.6% of the full-length in-sample KS comparisons. After all fits were frozen, it raised the mean cross-ticker KS pass rate from 83.5% under centered naive composition to 87.0% on the 2025 holdout and kept the median variance ratio at 0.97. The improvement was broad but not uniform: every method retained a lower tail of tickers with weak fits, and some hybrid fits deteriorated relative to matched-length training blocks. A high average pass rate is therefore not a guarantee for every asset.

Fitting the return generator directly to OLS residuals is a competitive alternative. Residual-fit methods avoid double counting because the market component has already been removed. The best such method, JumpHMM-on-residuals, had a slightly higher KS pass rate in sample, while the hybrid method had the higher rate in the holdout. The practical difference is reuse: the hybrid construction can use a generator already fitted to full returns, whereas the residual method requires another fit for each asset. Neither approach achieved nominal tail coverage in the short holdout. The hybrid method’s one-day left-tail 99% VaR exceedance rate was 1.67%, the same rounded rate as JumpHMM-on-residuals and above the 1% nominal rate.

Several design choices narrow this comparison. In sample, we used the observed SPY history as the market path. In the holdout experiment, we generated SPY from its frozen training-period model and reserved the observed 2025 path for scoring. The naive and hybrid methods shared the same JumpHMM draw for each ticker, while the residual methods used draws from their own fitted generators. We omitted the optional cross-sectional rank reorder and evaluated VaR for each ticker rather than for a portfolio. The variance correction also assumes $\text{Cov}(\tilde{g}_i^c, g_m) \approx 0$, which holds here because the asset and market paths were generated independently. A generator sampled from joint market history would require a covariance adjustment. In addition, the present construction adds only contemporaneous market dependence; it does not reproduce lead-lag effects or market-driven volatility clustering.

The sensitivity results mark the boundary of the observed evidence. Hybrid KS pass rates declined from 87.6% in the lowest β quartile to 65.9% in the highest because the market path determined more of the composed distribution as β increased. The clipping rule did not activate in the observed universe, where every ticker had $\rho_i < 0.90$, but it activated for most tickers after market volatility was doubled or tripled. Only two tracker assets used the R^2 -preserving branch. The 249-day holdout also had much less power than the 2,766-day training comparison. Matched-length training blocks partly address that difference, but one year of test data cannot provide a precise ranking from either pass rates or left-tail 99% VaR counts. The evidence therefore supports per-asset variance correction and distributional evaluation, not portfolio-level dependence fidelity.

The portfolio experiment shows why that distinction matters. Independent residual draws removed the sector and style covariance present in real assets (Booth and Fama, 1992). Daily rebalancing then converted the excess cross-sectional dispersion into a spurious return premium. This problem affected every method that generated residuals independently, including the block-bootstrap and GARCH variants. In a 20-asset target-weight example, the synthetic median annualized growth rate exceeded the realized 2025 return by about 18 percentage points. Moving from static weights to daily rebalancing changed the median by 27.73 percentage points, whereas adding the allocation signal changed it by only 0.13 percentage points (Booth and Fama, 1992; Bouchev et al., 2015; Markowitz, 1976). The apparent allocation gain was therefore almost entirely a diversification-return bias.

A joint residual model could address this limitation. A synchronized multivariate block bootstrap, residual factor model, or copula could restore cross-asset residual covariance. Multi-factor or lagged-factor versions could also capture dependence omitted by the current single-factor construction, with the variance correction retained as a compositional constraint. Per-asset jump calibration and stress tests involving highly volatile markets or leveraged assets would test the other transfer limits identified here. Separate univariate autoregressive or block-bootstrap models would not be enough: they can improve temporal behavior within each asset without creating dependence between assets.

The current location adjustment does not resolve this central limitation. We calibrated a constant shift in log-wealth to an external 8% annual long-run

return prior (Damodaran, 2025; Ibbotson and Duff and Phelps, 2024), not to the realized test-year result. The shift preserved path ranks and adjusted the return level, although it changed drawdown timing and depth as wealth accumulated. It did not restore the missing residual covariance. The variance correction is therefore necessary for combining full-return generators with a market factor, but it is not a complete dependence model. Portfolio-level use requires joint residual dynamics.

6. Conclusion

In this study, we developed a variance-corrected method for composing per-asset generators through a single-index model. Centering first prevented a full-return draw from shifting the calibrated intercept; the subsequent scale correction removed the variance inflation caused by adding the market factor. Across 423 non-market assets, it recovered the calibrated factor loading to a low median absolute error and produced heavy-tailed marginals. On 416 complete held-out asset histories from 2025, it improved the mean cross-ticker KS pass rate from 83.5% under centered naive composition to 87.0% and held the median variance ratio at 0.97. Its one-day left-tail 99% Value-at-Risk exceedance rate was 1.67%, the same rounded rate as JumpHMM-on-residuals but above the 1% nominal rate. Those residual models, fitted directly to the OLS residual series, required a separate fit for each asset and produced a similar marginal pass rate.

The per-asset component underpinning this construction was a hidden Markov marginal model with Laplace-quantile states, Student- t emissions, and direct transition counting. On SPY, its jump-duration variant reduced absolute-return autocorrelation error relative to the no-jump variant while keeping high KS and AD pass rates; GARCH reached a lower autocorrelation error but failed the distributional tests. The single-asset result therefore showed a measured tradeoff between distributional fit and volatility clustering, not uniform superiority. Taken together, these results separate two tasks that are easily conflated when synthesizing multi-asset market data: the marginal generator governs each asset’s distributional and temporal behavior, whereas the variance correction prevents the common market factor from double counting the generator’s total variance.

The current multi-asset implementation does not model cross-asset residual covariance. In a 20-asset allocator, independent residuals inflated the synthetic median daily-rebalanced return well above the realized 2025 outcome, almost entirely through the static-to-daily-rebalance diversification return rather than the allocation signal. A location-shift correction set to a documented 8% annual long-run prior adjusted the return level without restoring the missing covariance. A full joint residual model is needed before using these paths for portfolio-level return forecasts. The variance correction is a necessary compositional constraint rather than a complete dependence model. Portfolio-level synthetic market data would require combining it with joint residual dynamics.

Code and data availability

The experiment scripts, evaluation code (six-method comparison, left-tail Value-at-Risk coverage check, and bias correction), cached inputs and result summaries, and fitted per-asset marginals are publicly available. The underlying model-fitting library is released separately and pinned as a dependency of the experiment code. Identifying repository links have been withheld from this manuscript for double-anonymized review and are provided to the editorial office on the separate title page.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used OpenAI Codex to assist with language editing, consistency checks, and LaTeX formatting. The authors reviewed and edited all output and take full responsibility for the content of the publication.

References

- Aas, K., Czado, C., Frigessi, A., Bakken, H., 2009. Pair-copula constructions of multiple dependence. *Insurance: Mathematics and Economics* 44, 182–198.
- Anderson, T.W., Darling, D.A., 1952. Asymptotic theory of certain “goodness of fit” criteria based on stochastic processes. *The Annals of Mathematical Statistics* 23, 193–212.
- Arthur, W.B., 2021. Foundations of complexity economics. *Nature Reviews Physics* 3, 136–145. doi:10.1038/s42254-020-00273-3.
- Assefa, S.A., Dervovic, D., Mahfouz, M., Tillman, R.E., Reddy, P., Veloso, M., 2020. Generating synthetic data in finance: Opportunities, challenges and pitfalls, in: *Proceedings of the First ACM International Conference on AI in Finance*, pp. 1–8. doi:10.1145/3383455.3422554.
- Bachelier, L., 1900. Théorie de la spéculation. *Annales Scientifiques de l’École Normale Supérieure* 17, 21–86.
- Baum, L.E., Petrie, T., Soules, G., Weiss, N., 1970. A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *The Annals of Mathematical Statistics* 41, 164–171. doi:10.1214/aoms/1177697196.
- Bilmes, J.A., 1998. A Gentle Tutorial of the EM Algorithm and Its Application to Parameter Estimation for Gaussian Mixture and Hidden Markov Models. Technical Report TR-97-021. International Computer Science Institute.
- Black, F., Scholes, M., 1973. The pricing of options and corporate liabilities. *Journal of Political Economy* 81, 637–654. doi:10.1086/260062.

- Bollerslev, T., 1986. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics* 31, 307–327. doi:10.1016/0304-4076(86)90063-1.
- Booth, D.G., Fama, E.F., 1992. Diversification returns and asset contributions. *Financial Analysts Journal* 48, 26–32. doi:10.2469/faj.v48.n3.26.
- Bouchey, P., Nemtchinov, V., Wong, T.K.L., 2015. Volatility harvesting in theory and practice. *Journal of Wealth Management* 18, 89–100. doi:10.3905/jwm.2015.18.3.089.
- Bulla, J., Bulla, I., 2006. Stylized facts of financial time series and hidden semi-Markov models. *Computational Statistics & Data Analysis* 51, 2192–2209. doi:10.1016/j.csda.2006.07.021.
- Cherubini, U., Luciano, E., Vecchiato, W., 2004. *Copula Methods in Finance*. John Wiley & Sons.
- Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., Bengio, Y., 2014. Learning phrase representations using RNN encoder-decoder for statistical machine translation, in: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Association for Computational Linguistics. pp. 1724–1734. doi:10.3115/v1/D14-1179.
- Cont, R., 2001. Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance* 1, 223–236. doi:10.1080/713665670.
- Cont, R., Bouchaud, J.P., 2000. Herd behavior and aggregate fluctuations in financial markets. *Macroeconomic Dynamics* 4, 170–196. doi:10.1017/S1365100500015029.
- Damodaran, A., 2025. Equity risk premiums: Determinants, estimation and implications. NYU Stern School of Business working paper, updated annually. <https://pages.stern.nyu.edu/~adamodar/>.
- Dias, J.G., Vermunt, J.K., Ramos, S., 2010. Mixture hidden Markov models in finance research, in: *Advances in Data Analysis, Data Handling and Business Intelligence*, Springer. pp. 451–459. doi:10.1007/978-3-642-01044-6_41.
- Embrechts, P., McNeil, A.J., Straumann, D., 2002. Correlation and dependence in risk management: Properties and pitfalls, in: *Risk Management: Value at Risk and Beyond*. Cambridge University Press, pp. 176–223. doi:10.1017/CB09780511615337.008.
- Engle, R.F., 1982. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica* 50, 987–1007.
- Fama, E.F., 1970. Efficient capital markets: A review of theory and empirical work. *The Journal of Finance* 25, 383–417. doi:10.2307/2325486.

- Fama, E.F., French, K.R., 1993. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics* 33, 3–56. doi:10.1016/0304-405X(93)90023-5.
- Farmer, J.D., 2025. Quantitative agent-based models: A promising alternative for macroeconomics. *Oxford Review of Economic Policy* 41, 571–590. doi:10.1093/oxrep/graf027.
- Farmer, J.D., Joshi, S., 2002. The price dynamics of common trading strategies. *Journal of Economic Behavior & Organization* 49, 149–171. doi:10.1016/S0167-2681(02)00065-3.
- Glasserman, P., 2003. *Monte Carlo Methods in Financial Engineering*. Springer. doi:10.1007/978-0-387-21617-1.
- Grinstead, C.M., Snell, J.L., 1997. *Introduction to Probability*. American Mathematical Society.
- Hamilton, J.D., 1989. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* 57, 357–384. doi:10.2307/1912559.
- Hamilton, J.D., 2018. Regime switching models. *The New Palgrave Dictionary of Economics* , 11421–11426doi:10.1057/978-1-349-95189-5_2459.
- Heston, S.L., 1993. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies* 6, 327–343. doi:10.1093/rfs/6.2.327.
- Ibbotson, R.G., Duff and Phelps, 2024. *SBBI Yearbook: Stocks, Bonds, Bills, and Inflation. Annual ed.*, Duff and Phelps / Kroll.
- Jordon, J., Szpruch, L., Houssiau, F., Bottarelli, M., Cherubin, G., Maple, C., Cohen, S.N., Weller, A., 2022. Synthetic data: What, why and how? *arXiv preprint arXiv:2205.03257* .
- Kim, S., Shephard, N., Chib, S., 1998. Stochastic volatility: Likelihood inference and comparison with ARCH models. *The Review of Economic Studies* 65, 361–393.
- Kolmogorov, A.N., 1933. Sulla determinazione empirica di una legge di distribuzione. *Giornale dell'Istituto Italiano degli Attuari* 4, 83–91.
- Kotz, S., Kozubowski, T.J., Podgórski, K., 2001. *The Laplace Distribution and Generalizations: A Revisit with Applications to Communications, Economics, Engineering, and Finance*. Birkhauser.
- Kou, S.G., 2002. A jump-diffusion model for option pricing. *Management Science* 48, 1086–1101. doi:10.1287/mnsc.48.8.1086.166.

- Kwon, S., Lee, Y., 2024. Can GANs learn the stylized facts of financial time series?, in: Proceedings of the 5th ACM International Conference on AI in Finance, pp. 126–133. doi:10.1145/3677052.3698661.
- LeBaron, B., 2006. Agent-based computational finance, in: Tesfatsion, L., Judd, K.L. (Eds.), Handbook of Computational Economics. Elsevier. volume 2, pp. 1187–1233. doi:10.1016/S1574-0021(05)02024-1.
- Lux, T., Marchesi, M., 1999. Scaling and criticality in a stochastic multi-agent model of a financial market. *Nature* 397, 498–500. doi:10.1038/17290.
- Mandelbrot, B., 1963. The variation of certain speculative prices. *The Journal of Business* 36, 394–419.
- Markowitz, H.M., 1976. Investment for the long run: New evidence for an old rule. *Journal of Finance* 31, 1273–1286. doi:10.1111/j.1540-6261.1976.tb03213.x.
- Merton, R.C., 1976. Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics* 3, 125–144. doi:10.1016/0304-405X(76)90022-2.
- Nguyen, N., 2018. Hidden Markov model for stock trading. *International Journal of Financial Studies* 6, 36. doi:10.3390/ijfs6020036.
- Osborne, M.F.M., 1959. Brownian motion in the stock market. *Operations Research* 7, 145–173.
- Politis, D.N., Romano, J.P., 1994. The stationary bootstrap. *Journal of the American Statistical Association* 89, 1303–1313. doi:10.1080/01621459.1994.10476870.
- Rabiner, L., Juang, B., 1986. An introduction to hidden Markov models. *IEEE ASSP Magazine* 3, 4–16. doi:10.1109/MASSP.1986.1165342.
- Ross, S.A., 1976. The arbitrage theory of capital asset pricing. *Journal of Economic Theory* 13, 341–360. doi:10.1016/0022-0531(76)90046-6.
- Rossi, A., Gallo, G.M., 2006. Volatility estimation via hidden Markov models. *Journal of Empirical Finance* 13, 203–230. doi:10.1016/j.jempfin.2005.09.003.
- Rydén, T., Teräsvirta, T., Åsbrink, S., 1998. Stylized facts of daily return series and the hidden Markov model. *Journal of Applied Econometrics* 13, 217–244. doi:10.1002/(SICI)1099-1255(199805/06)13:3<217::AID-JAE476>3.0.CO;2-V.
- Samuelson, P.A., 1965. Rational theory of warrant pricing. *Industrial Management Review* 6, 13–32.

- Schwert, G.W., 1989. Why does stock market volatility change over time? The Journal of Finance 44, 1115–1153. doi:10.1111/j.1540-6261.1989.tb02647.x.
- Sharpe, W.F., 1963. A simplified model for portfolio analysis. Management Science 9, 277–293. doi:10.1287/mnsc.9.2.277.
- Smirnov, N., 1948. Table for estimating the goodness of fit of empirical distributions. The Annals of Mathematical Statistics 19, 279–281.
- Stein, E.M., Stein, J.C., 1991. Stock price distributions with stochastic volatility: An analytic approach. The Review of Financial Studies 4, 727–752.
- Stenger, M., Bauer, A., Prantl, T., Leppich, R., Hudson, N., Chard, K., Foster, I., Kounev, S., 2024. Thinking in categories: A survey on assessing the quality for time series synthesis. Journal of Data and Information Quality 16, 1–32. doi:10.1145/3666006.
- Takahashi, S., Chen, Y., Tanaka-Ishii, K., 2019. Modeling financial time-series with generative adversarial networks. Physica A: Statistical Mechanics and Its Applications 527, 121261. doi:10.1016/j.physa.2019.121261.
- Toth, D., Jones, B., 2019. Against the norm: Modeling daily stock returns with the Laplace distribution. arXiv preprint arXiv:1906.10325 .
- Yoon, J., Jarrett, D., van der Schaar, M., 2019. Time-series generative adversarial networks, in: Advances in Neural Information Processing Systems, pp. 5509–5519.